

Integrals Expressed Through Special Functions

Topic.

Special functions, integration by substitution, integration by parts.

Statement.

There are integrals that cannot be expressed through elementary functions but often appear in applications. Examples include the integral sine

$$\text{Si}(x) = \int \frac{\sin x}{x} dx$$

and the logarithmic integral

$$\text{li}(x) = \int \frac{dx}{\ln x}, \quad x \in (0, 1).$$

Express the following integrals through the integral sine, the logarithmic integral, and elementary functions:

$$\int \frac{\sin(x^2)}{x} dx$$

and

$$\int \frac{dx}{\ln^2 x}, \quad x \in (0, 1).$$

Solution.

Denote

$$\text{Si}(x) = \int \frac{\sin x}{x} dx,$$

and

$$\text{li}(x) = \int \frac{dx}{\ln x}.$$

First, consider

$$\int \frac{\sin(x^2)}{x} dx.$$

Make the substitution

$$t = x^2.$$

Then

$$dt = 2x \, dx.$$

Therefore,

$$\frac{dt}{2x^2} = \frac{2x \, dx}{2x^2} = \frac{dx}{x}.$$

Since

$$x^2 = t,$$

we get

$$\frac{dx}{x} = \frac{dt}{2t}.$$

Hence,

$$\int \frac{\sin(x^2)}{x} \, dx = \int \sin t \cdot \frac{dt}{2t} = \frac{1}{2} \int \frac{\sin t}{t} \, dt.$$

By the definition of the integral sine,

$$\frac{1}{2} \int \frac{\sin t}{t} \, dt = \frac{1}{2} \operatorname{Si}(t) + C.$$

Returning to x , we obtain

$$\int \frac{\sin(x^2)}{x} \, dx = \frac{1}{2} \operatorname{Si}(x^2) + C.$$

Now consider

$$\int \frac{dx}{\ln^2 x}, \quad x \in (0, 1).$$

Compute the derivative of

$$\frac{x}{\ln x}.$$

We have

$$\left(\frac{x}{\ln x} \right)' = \frac{\ln x - x \cdot \frac{1}{x}}{\ln^2 x} = \frac{\ln x - 1}{\ln^2 x}.$$

Therefore,

$$\left(\frac{x}{\ln x}\right)' = \frac{1}{\ln x} - \frac{1}{\ln^2 x}.$$

Hence,

$$\frac{1}{\ln^2 x} = \frac{1}{\ln x} - \left(\frac{x}{\ln x}\right)'.$$

Then

$$\int \frac{dx}{\ln^2 x} = \int \frac{dx}{\ln x} - \int \left(\frac{x}{\ln x}\right)' dx.$$

By the definition of the logarithmic integral,

$$\int \frac{dx}{\ln x} = \text{li}(x).$$

Thus,

$$\int \frac{dx}{\ln^2 x} = \text{li}(x) - \frac{x}{\ln x} + C, \quad x \in (0, 1).$$